

IJASC 26-2-x

Detection of Cardiac Structural Abnormalities from Temporal Chest X-Ray Sequences Using Deep Learning

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Abstract

Cardiovascular diseases remains one of the most life threatening issues in the world, with over 17 million deaths each year. while echocardiography remains the most effective method of attending to these deaseses, it remains expensive and unavailable in resource insufficient areas . This study examines whether analyzing series of chest X-rays taken over time provides more diagnostic value compare to looking at a single image. Using the CheXchoNet dataset, which contains 71,589 chest X-rays from 24,689 patients, each paired with echocardiography data, we trained a transformer-based temporal model to read sequences of up to four X-rays and compared it against a standard single-image baseline. The temporal model outperform the baseline on all three conditions: severe left ventricular hypertrophy (SLVH), dilated left ventricle (DLV), and a composite cardiac label. Mean AUC was 0.7931 versus 0.7781 (+1.93%). DLV showed the strongest gain at +2.04% (AUC 0.8213 vs 0.8049, $p=0.0625$), and this hold across every validation fold. Sequential X-ray analysis captures how the heart changes over time, something a experienced single-image model simply cannot do.

Keywords: cardiac imaging, chest X-ray, deep learning, dilated cardiomyopathy, left ventricular hypertrophy, temporal modeling.

Manuscript Received: xxxxxx / Revised: xxxxxx / Accepted: xxxxxx

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1. INTRODUCTION

Heart disease is the world's leading killer. Naturally doctors compare two or more x-rays to determine the presence, type, nature or progress of the disease. That comparison over time is often what separates a routine finding from a serious one [8]. This differs with most widely used AI systems that only check one and most recent x-ray image.

This paper addresses the question that if you have a model trained from a sequence of chest x-rays taken over time from the same patient over a time period, will it detect the disease more accurately than the model that takes the last scan?. In this paper we focus on two structural heart disease (SLVH) and (DLV). These conditions usually develop slowly and they sometimes end up undetected as they leave traces that accumulate across multiple scans[6].

We built a hybrid CNN-Transformer model to read up to four sequential X-rays per patient and compared it against a ResNet-18 single-image baseline. Both were tested on the CheXchoNet dataset [21], which pairs chest X-rays with echocardiography measurements, the accepted clinical standard for heart structure assessment.

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2. CARDIAC CONDITIONS AND RELATED WORK

2.1 Target Conditions

SLVH refers to when the left ventricle has a thickened wall (typically >13 - 15 mm). They are most commonly seen in long-term hypertension, aortic stenosis and hypertrophic cardiomyopathy [13][14]. Since thickening occurs gradually, early signs on a sole X-ray are easy to miss.

DLV is the opposite: the left ventricle dilates and walls thin, with end-diastolic diameter > 55 to 60 mm. These can be caused by a viral or alcohol infection of the heart muscle, certain cancer drugs and genetic factors [15]. DLV is the leading cause of heart transplants for both adults and children.

2.2 Related Work

Deep learning has achieved expert-level performance on a number of radiology tasks, such as the detection of pneumonia [10], screening for diabetic retinopathy [16] and classification of multi-label chest X-ray images [7]. In particular for LV abnormalities Bhav et al. [20] a CNN trained on single X-rays yielded AUROC of 0.777, outperforming the performance of 15 radiologists, though still only provided a single image per patient.

Temporal approaches remain rare. Santeramo et al. [18] demonstrated that using time-modulated LSTMs helps to better detect radiological abnormalities from sequential chest X-rays whilst explicitly dealing with irregular temporal gaps between scans.

3. METHODOLOGY

3.1 Dataset

We worked with the CheXchoNet dataset [21]: 71 589 chest X-ray studies of 24 689 patients of Columbia University Irving Medical Center. Echocardiography measurements (IVSd, LVIDd, LVPWd) where three binary classification labels were obtained. Table 1 summarizes the class distribution. Imbalance is acute, especially in the case of DLV.

Table 1. Class distribution in the CheXchoNet dataset.

Condition	Positive Cases	Negative Cases	Prevalence	Imbalance Ratio
SLVH	1,789 (9.08%)	17,922 (90.92%)	9.08%	10.02:1
DLV	906 (4.60%)	18,805 (95.40%)	4.60%	20.76:1
Composite	2,503 (12.69%)	17,208 (87.31%)	12.69%	6.88:1

3.2 Preprocessing

The X-rays of each patient were placed in order of time and the four latest ones were taken. In case there were less than four, the sequence was padded with a repeated image. All images were scaled to 224x224 pixels, turned to 3-channel RGB and normalized with ImageNet mean and standard deviation. Metadata (age, sex, IVSd, LVPWd, LVIDd) of the patient was coded using a 12-dimensional feature vector and combined with the image features at the classification step.

3.3 Model Architecture

Figure 1 shows the full model pipeline. The system operates on a single-image path as well as a temporal sequence path using a common ResNet-18 CNN encoder [22], which produces a 256-dimensional feature-vector per image. In the case of the temporal path, these vectors are inputted to a 2-layer Transformer encoder [12] (4 attention heads, FFN dimension 1024, dropout 0.2) positional embeddings that learn the scan order. A time-step attention pooling layer summarizes thereafter. In the case of the single-image path, the sequence is squeezed directly to the 256-dim representation. The two paths converge at a concatenation phase where the 256-dimensional image features are concatenated with the 12-dimensional patient metadata vector, resulting in a 268-dimensional input to the common classification head. The final prediction of each condition is a 3-layer MLP (268 to 512 to 256 to 3).

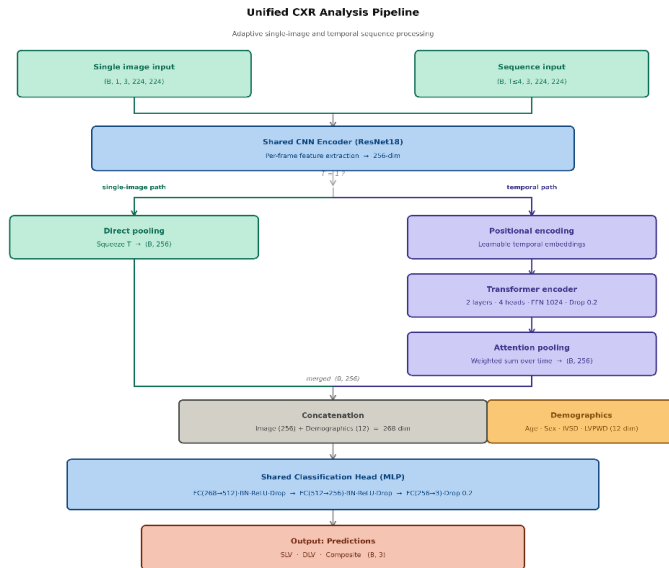


Figure 1. Model pipeline showing both the single-image path (left) and temporal sequence path (right), merging at concatenation before the shared classification head.

3.4 Training Details

Given class imbalance as high as 20.76:1, we used Adaptive Weighted Binary Cross-Entropy (AW-BCE) loss, which scales the positive-class contribution by the inverse of its prevalence. The AdamW optimizer was used with a learning rate of 2×10^{-4} , gradient accumulation over 2 steps (effective batch size 256), and early stopping with patience of 5 epochs. All results are from stratified 5-fold cross-validation [23].

4. RESULTS

The temporal model outperformed the single-image baseline on every condition in every fold. Table 2 shows the full comparison. Statistical significance was assessed using the Wilcoxon signed-rank test [24] on the five paired fold results.

Table 2. Performance comparison: temporal vs. single-image model (5-fold cross-validation). * $p < 0.05$.

Metric	Temporal AUC	Single AUC	Δ (%)	p-value	Wins
AUC SLVH	0.7637	0.7552	+1.13	0.4375	4/5
AUC DLV	0.8213	0.8049	+2.04	0.0625	5/5
AUC Composite	0.7896	0.7743	+1.98	0.1250	5/5
AUC Average	0.7931	0.7781	+1.93	0.0286*	5/5

The overall AUC improvement of +1.93% ($p=0.0286$) is statistically significant. DLV had the largest gain (+2.04%), and the temporal model won all five folds for DLV and the composite condition. For SLVH the improvement was smaller (+1.13%, $p=0.4375$) but the temporal model still came out ahead in 4 of 5 folds.

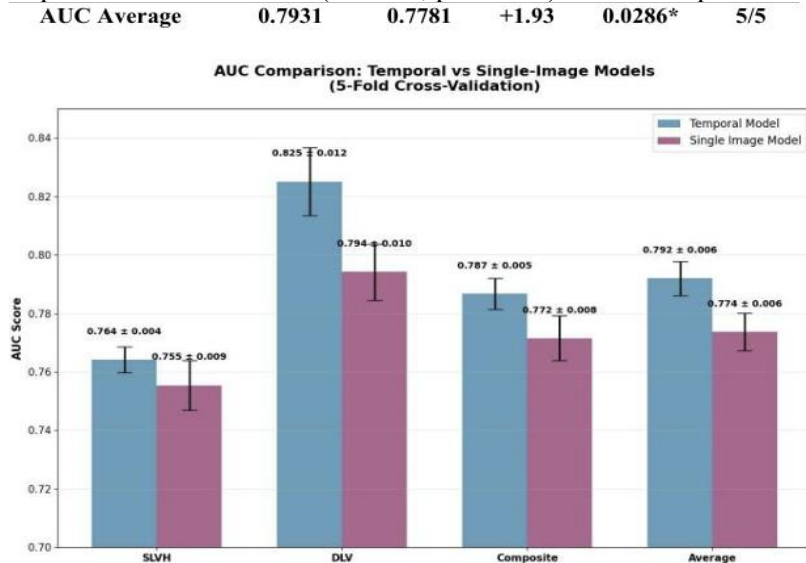


Figure 2. AUC comparison across all three conditions and the overall average. Error bars show standard deviation across folds. The temporal model (blue) consistently exceeds the single-image model (pink).

The greatest gain was on DLV, which corresponds to the actual development of dilated cardiomyopathy. The enlarges gradually and gradually, over months or years, ventricle. A model that has access to past scans is able to detect. that drift. One image is not enough to have a model. Transformer self-attention mechanism is an effective learning mechanism. to concentrate on which historical scans convey the greatest signal to the present prediction, similar to a skilled. radiologist searching through previous films.

5. DISCUSSION

This finding demonstrates that monitoring of changes in chest X-rays of a patient across time provides a real diagnostic value, which cannot be duplicated by a single-image model. The returns are small but steady in direction and the ideal consistency of the fold in DLV is the type of strength that is important in a clinical environment.

Practically, hospitals already store and accumulate the chest X-rays with time. There is no requirement of new equipment. This model has the potential to be executed on existing PACS archives and thus it is highly applicable to resource constrained environments that lack echocardiography.

5.1 Limitations

All data came from one institution, which limits generalizability. The time gap between scans was variable, introducing noise the model cannot fully account for. DLV prevalence was only 4.6%, constraining statistical power even with loss reweighting. Also, AUC gains did not always translate into improved precision-recall performance [24], the more informative measure when classes are highly imbalanced.

5.2 Future Work

Priorities include multi-center validation, self-supervised pre-training on unlabeled longitudinal X-ray data, federated learning [25] for privacy-preserving collaboration across hospitals, and prospective clinical trials to assess real-world impact.

6. CONCLUSION

Reading sequences of chest X-rays over time improves the detection of structural heart abnormalities compared to single-image analysis. Our CNN-Transformer model achieved a mean AUC of 0.7931 versus 0.7781 for the single-image baseline (+1.93%, $p=0.0286$). The improvement was strongest for DLV (+2.04%, 5/5 folds), where the gradual nature of disease progression makes temporal information especially valuable. Because the model operates on X-rays that hospitals already collect, it could support cardiac screening in settings where echocardiography is inaccessible.

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